

Technical Architecture

How the CHT platform is built and run: frontend, backend, database, authentication, APIs, storage, third-party services, environments and external dependencies.



At a glance

- React single-page app on **S3 + CloudFront**, with `/api` served from the same domain through an ALB.
- **NestJS** API and a **Python** worker on **ECS Fargate**, deployed by Terraform from GitHub Actions.
- **PostgreSQL 15**: Aurora Global in production (us-east-1 writer, us-east-2 reader), RDS on dev.
- **Amazon Cognito** for member and admin sign-in (MFA, Google) and for service-to-service tokens with Content Hub.
- Two environments in one AWS account: **Dev** (`devapp.communityhealth.media`) and **Platform / production** (`testapp.communityhealth.media` , alias `app.communityhealth.media`).

Document	CHT Platform Technical Architecture
Version	1.0
Date	September 29, 2026
Status	Draft for review
Owner	Uche Aduaka (CHT Platform)
Reviewers	CHT engineering, Content Hub engineering, Security, CHM leadership
Source of truth	<code>cht-platform-tool</code> repository: <code>frontend/</code> , <code>backend/</code> , <code>worker/</code> , <code>infrastructure/terraform/</code> , <code>.github/workflows/</code>

Contents

1	Description	7	Storage
2	Frontend	8	Third-party services
3	Backend	9	Production environment
4	Database	10	Dev environment
5	Authentication	11	External dependencies
6	APIs	12	Open items

Status: **Live** running today, **Transition** being migrated or partly rolled out, **Planned** designed but not yet deployed.

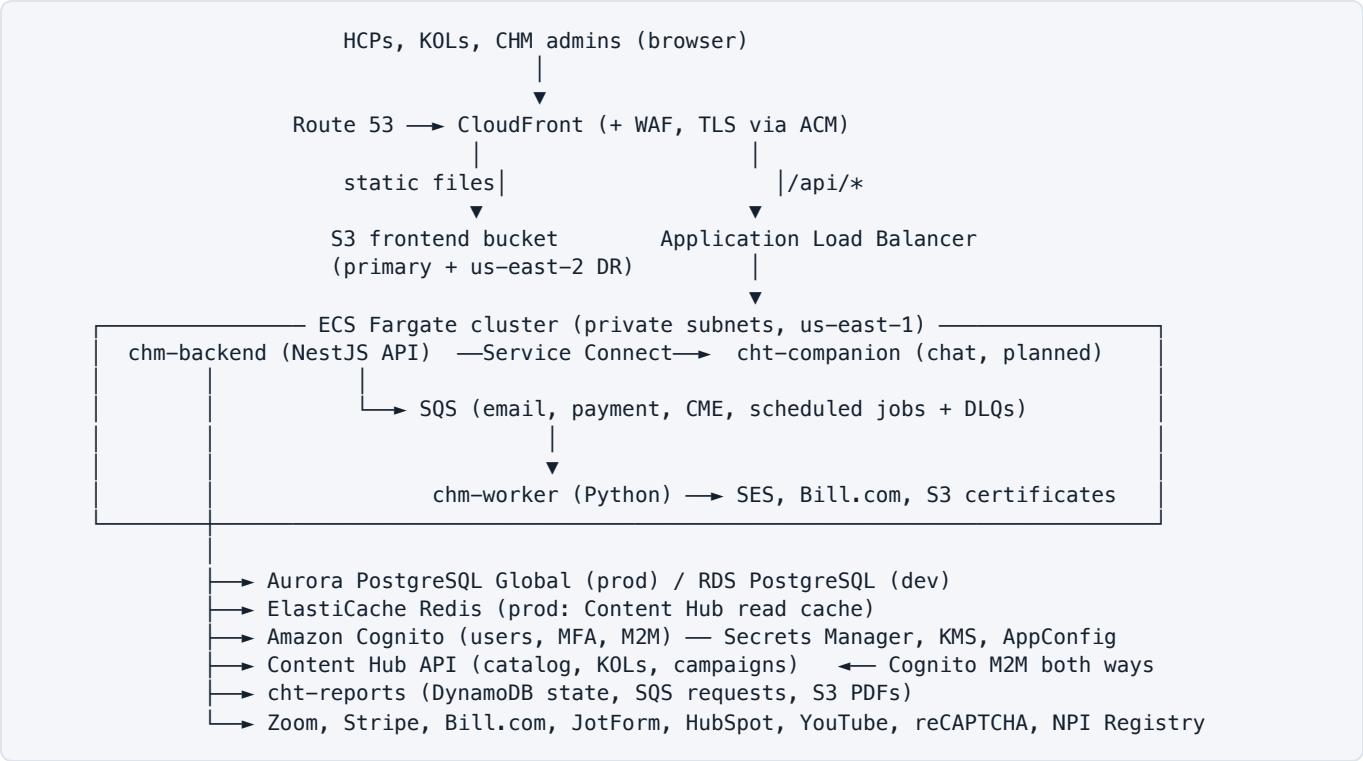
1 Description

The CHT platform is the web product behind Community Health Media. It serves three audiences from one codebase and one domain per environment:

- **Public site:** marketing pages, the clinical conversation catalog, the KOL network, podcasts and upcoming programs.
- **Member app** (/app): HCPs register for webinars and office hours, attend in Zoom, complete surveys, get paid honoraria, and watch content.
- **Admin console** (/admin): CHM staff schedule programs, approve registrations, manage surveys, users and payments, review audit logs and Zoom recordings, and generate sponsor reports.

The platform owns identity, programs, registrations, attendance, surveys and payments. Catalog content, KOL data and campaign analytics come from the separate **Content Hub** service. Executive PDF reports are produced by the separate **cht-reports** pipeline.

System overview



2 Frontend

Aspect	Detail
Framework	React 19 single-page app, TypeScript, built with Vite 7
Routing	React Router 7 with lazy-loaded route chunks; public, /app (member) and /admin trees with auth and role guards
Data	TanStack React Query 5 over an Axios client; cookies sent with every request (withCredentials); polling for long-running jobs such as reports
UI	Tailwind CSS 3, shared kit in components/ui (Button, Field, Chip, SegmentedControl) and admin patterns (section cards, alerts, status badges); Lucide icons; Recharts for analytics
Payments UI	Stripe Connect embedded components (@stripe/connect-js) for payout onboarding
Zoom	Meeting SDK Client View 5.1.4 loaded from the Zoom CDN inside a same-origin iframe (/zoom-embed.html). CloudFront sends Cross-Origin-Opener-Policy: same-origin and Cross-Origin-Embedder-Policy: credentialless so live video works.
Bot protection	Google reCAPTCHA v3 on join and login when keys are configured
Quality	ESLint (React Compiler rules), tsc , Vitest unit tests; all run in CI before deploy

Aspect	Detail
Hosting	Static build synced to S3 and served by CloudFront; hashed assets cached long-term, <code>index.html</code> and <code>zoom-embed.html</code> uploaded separately; CloudFront invalidation after each deploy

Because the SPA and API share one origin, session cookies are first-party and no cross-origin CORS is needed for the browser. The API also exposes `Content-Disposition` so downloaded files (for example report PDFs) keep friendly names.

3 Backend

3.1 API service (chm-backend)

Aspect	Detail
Runtime	Node.js 20 (Alpine container), NestJS 11, TypeScript
Data access	Prisma 5.22 ORM for PostgreSQL; AWS SDK v3 for DynamoDB, S3, SQS and Cognito; ioredis for the cache
Structure	Feature modules: admin, auth, campaigns, catalog, companion, contact, content-hub, dashboard, email, export, hubspot, internal, invites, jotform, kol-network, outbound-sync, payments, podcasts, programs, reports, surveys, webinars, zoom-recordings
Conventions	Global <code>/api</code> prefix; DTO validation on every input; guards for session, roles (ADMIN) and M2M scopes; admin actions written to the audit log
Security	Helmet headers, HSTS in production, <code>trust proxy</code> behind the ALB, rate-limited auth with account lockout, Swagger docs (<code>/api/docs</code>) gated to admin sessions
Logging	Structured JSON logs (pino) to CloudWatch; tokens, secrets and Authorization headers are never logged
Health	<code>GET /health</code> (liveness), <code>GET /health/ready</code> (database; used by the ALB and deploy gates), <code>GET /actuator/info</code> (region, image tag)
Config	Environment variables validated at start-up; secrets injected from Secrets Manager into the ECS task; feature flags through AWS AppConfig

3.2 Worker (chm-worker)

Aspect	Detail
Runtime	Python 3.11 (slim container), boto3, SQLAlchemy with psycopg2, ReportLab and Pillow
Consumers	SQS queues for email (SES), payment (honorarium payouts) and CME (certificate PDFs to S3), plus a scheduled jobs queue
Reliability	Each queue has a dead-letter queue with a CloudWatch alarm; health check script for ECS

Aspect	Detail
Schedules	EventBridge rules: session reminders (every 12 hours on dev, every 6 hours in production) and, in production, a Bill.com MFA reminder every 20 days

3.3 Key flows

1. **Program to payment.** Admin schedules a program (Zoom meeting or webinar created by API) → HCP registers and is approved → Zoom webhooks and in-app SDK events record attendance → HCP submits the post-event survey → backend checks eligibility and queues a payment → worker pays out and updates status.
2. **Executive report.** Admin requests a report → backend writes state to DynamoDB, sends a message to the report queue and returns 202 → cht-reports pulls Content Hub and Platform data (via the M2M export endpoint), renders a PDF to S3 → backend streams the PDF to the admin without exposing the S3 location.
3. **Catalog.** Backend reads Content Hub public and admin APIs with an M2M Bearer token, caches results in Redis (production, 1 hour), and serves them to the SPA.

4 Database

4.1 PostgreSQL (system of record)

	Production (platform)	Dev
Engine	Aurora PostgreSQL Global 15.17	RDS PostgreSQL 15.17
Topology	Writer in us-east-1, reader in us-east-2 (managed failover)	Single instance, single AZ
Size	db.r6g.large	db.t3.micro, 20 GB
Backups	7-day retention	1-day retention
Encryption	At rest with KMS; TLS in transit; private subnets only; AWS Config rule checks RDS encryption	

The schema is managed by Prisma migrations (51 to date), applied automatically during each deploy through ECS Exec on the backend task. It has 24 models, grouped as follows:

Area	Models
Identity	User, Session, KolDirectoryVisibility
Programs	Program, OfficeHoursSlot, ProgramRegistration, ProgramEnrollment, RegistrationInvite, ProgramFormLink
Zoom	WebinarParticipantEvent, ZoomRecordingSession, ZoomRecordingFile, ZoomSyncJob, ZoomAttendanceImportJob, ZoomAttendanceParticipant
Surveys	Survey, SurveyResponse, FormJotformProgress
Payments and CME	Payment, CMECredit

Area	Models
Content and analytics	Video , VideoView , AnalyticsEvent
Governance	AdminAuditLog

4.2 Other data stores

Store	Used for	Owner
DynamoDB	Report state (cht-{env}-report-state , partition key campaign_id , sort key report_id , report_id-index); per-campaign lock rows prevent duplicate generation	cht-reports (platform has scoped access)
ElastiCache Redis	Read cache for Content Hub responses (production, cache.t3.medium)	Platform
Companion DB	PostgreSQL with pgvector for chat retrieval	cht-companion Planned

5 Authentication

5.1 Members and admins

Aspect	Detail
Identity provider	Amazon Cognito user pool per environment (Essentials tier), Hosted UI domains chm-dev and chm-platform
Sign-in methods	Email and password (POST /api/auth/cognito/login); Google through Cognito Hosted UI (/auth/callback)
MFA	Optional at the pool level with TOTP and SMS; the app prompts all users to enroll (/mfa/setup). Plan: switch to required once users have enrolled.
Sessions	Backend exchanges Cognito tokens for its own session: httpOnly, secure cookie backed by the Session table; revoked on logout and password change
Authorization	Roles on User (member, KOL, ADMIN); route guards in the SPA and enforced server-side on every API
Protection	Cognito WAF, CloudFront WAF, reCAPTCHA v3, login lockout, responses that don't reveal whether an email exists
Resilience	Cognito multi-region replica in us-east-2 (production and dev); email sent from noreply@communityhealth.media via SES
Profile checks	NPI validated against the public NPI Registry during profile completion

5.2 Service-to-service

Direction	Mechanism	Scope / secret
Content Hub → Platform	Cognito client credentials, Bearer token	platform/export.read (report input packet), platform/cache.clear
Platform → Content Hub	Cognito client credentials (client cht-platform-m2m-{env}), Bearer token cached until expiry	Hub resource server scopes (public and admin APIs)
Platform → Companion	ECS Service Connect (private) plus internal shared secret	COMPANION_INTERNAL_SECRET
Vendors → Platform	Webhook signatures (HMAC, Stripe-Signature)	Zoom, Stripe, Stripe Connect, Bill.com, JotForm webhook secrets

6 APIs

All endpoints are served under /api on the environment domain. There are 29 controllers; the main groups are below.

Group	Base path	Access	Purpose
Auth	/api/auth	Public / session	Login, join, Google callback, MFA, password reset, session
Catalog and content	/api/catalog , /api/kol-network , /api/podcasts	Public	Clips, playlists, KOL directory, podcast episodes
Programs	/api/programs , /api/webinars , /api/office-hours , /api/invites	Public / member	Listings, registration, Meeting SDK auth, SDK attendance
Surveys	/api/surveys , /api/jotform	Member	Native surveys, drafts, submissions, legacy JotForm
Payments	/api/payments	Member	Earnings, payout onboarding, payment history
Member dashboard	/api/dashboard	Member	Home data, notifications

Group	Base path	Access	Purpose
Admin	/api/admin/*	ADMIN	Programs, approvals, users, HCP explorer, surveys, payments, audit log, campaigns, KOL curation, Zoom recordings, Content Hub campaigns and program links
Reports	/api/reports	ADMIN	Generate (202), list, poll, regenerate, streamed PDF download
Export	/api/export	M2M	Report input packet for Content Hub / cht-reports
Internal	/api/internal/cache	M2M	Cache clear from Content Hub
Webhooks	/api/webhooks/{zoom,stripe,bill,jotform}	Signed	Vendor events
Other	/api/contact , companion chat routes	Public / member	Contact form, chat proxy
Ops	/health , /health/ready , /actuator/info , /api/docs	Public / ADMIN	Health, build info, OpenAPI (admin only)

7 Storage

Store	Contents	Access
S3 frontend	SPA build (primary bucket plus a synced us-east-2 DR bucket)	CloudFront origin access only
S3 certificates	CME certificate PDFs written by the worker	Backend and worker task roles
S3 session assets	Program and session images and files	Backend task role; public URL base for approved assets
S3 report artifacts	Executive report PDFs under reports/... (dev: cht-reports-dev-artifacts)	Owned by cht-reports; platform reads via bucket policy and streams to admins. Only reports/ keys and PDF content are served.
S3 audit buckets	CloudTrail and AWS Config logs	Security tooling only
ECR	Backend and worker images with lifecycle rules; replicated to us-east-2 in production	CI and ECS

Store	Contents	Access
Secrets Manager	Database URL, vendor keys, Cognito M2M client secrets; replicated to us-east-2 in production	ECS task roles; set by Terraform from GitHub Environment secrets
KMS	Keys for database, secrets and bucket encryption	Service roles

All buckets block public access; an AWS Config rule checks for public-read buckets.

8 Third-party services

Service	Used for	Integration	Status
Zoom	Create webinars and meetings, attendance, in-app join, cloud recordings and transcripts	Server-to-Server OAuth, Meeting SDK JWT, signed webhooks	Live
Stripe Connect	Honorarium payouts (ACH), tax detail collection in onboarding	Secret key, embedded onboarding, Connect webhooks	Transition
Bill.com	Legacy honorarium payouts; disabled wherever Stripe is configured	Connect v3 API from the worker, webhooks	Transition
JotForm	Legacy surveys for older programs	API key, webhooks	Transition
HubSpot	Campaign and funnel data	Private app access token	Live
YouTube Data API	Catalog playlists and podcast episodes	API key	Live
Google	Sign-in (through Cognito) and reCAPTCHA v3	OAuth IdP, site and secret keys	Live
NPI Registry (CMS)	Validate HCP NPI numbers	Public API	Live
Amazon Bedrock	LLM for the member chat companion	Via cht-companion	Planned

AWS managed services used directly: CloudFront, Route 53, ACM, WAF, ALB, ECS Fargate, ECR, Aurora/RDS, DynamoDB, ElastiCache, SQS, SES, EventBridge, SNS, Secrets Manager, KMS, AppConfig, CloudWatch, CloudTrail, AWS Config and GuardDuty.

9 Production environment

Aspect	Detail
Name	Platform (cht-platform), GitHub Environment platform
Domains	testapp.communityhealth.media , alias app.communityhealth.media

Aspect	Detail
Regions	Primary us-east-1; disaster recovery standby in us-east-2
Compute	Backend: 2 tasks (0.5 vCPU, 1 GB). Worker: 2 tasks (0.5 vCPU, 1 GB).
Database	Aurora Global <code>db.r6g.large</code> , writer us-east-1, reader us-east-2, 7-day backups
Cache	ElastiCache Redis <code>cache.t3.medium</code>
Network	VPC with public and private subnets across AZs; one NAT gateway per AZ
Content Hub	<code>contenthub.communityhealth.media</code>
Disaster recovery	CloudFront origin failover to the us-east-2 ALB; Aurora managed failover; replicated secrets, ECR images and frontend bucket; Cognito replica. DR compute is deployed separately (<code>deploy-secondary.sh</code>); API traffic is not routed there by default.
Monitoring	CloudWatch alarms (ALB 5xx, application errors, ECS and database CPU, primary down, every DLQ) to SNS email; GuardDuty findings; CloudTrail; AWS Config rules
Deploy	<code>deploy-prod.yml</code> : runs when a release or hotfix PR merges to <code>main</code> , or manually. Tests, image build to ECR, Terraform plan, manual approval, apply, migrations, frontend sync, health gate. Deploys are queued, never cancelled. <code>rollback.yml</code> redeploys a known-good image.

10 Dev environment

Aspect	Detail
Name	Dev (<code>cht-dev</code>), GitHub Environment <code>dev</code>
Domain	<code>devapp.communityhealth.media</code> (API at <code>/api</code>)
Region	us-east-1 (single region; no API failover)
Compute	Backend and worker: 1 task each (0.25 vCPU, 512 MB), autoscale to 2
Database	RDS PostgreSQL <code>db.t3.micro</code> , single AZ, 1-day backups
Network	Single shared NAT gateway to reduce cost
Content Hub	<code>devhub.communityhealth.media</code>
Reports	DynamoDB <code>cht-dev-report-state</code> , SQS <code>cht-dev-report-requests</code> , S3 <code>cht-reports-dev-artifacts</code>
Companion	Service Connect namespace <code>cht-dev.local</code> , <code>http://cht-companion:8080</code>
Transcripts	Zoom VTT uploads trigger the Content Hub dev ingest Lambda
Deploy	<code>deploy-dev.yml</code> : runs on every push to <code>feature/**</code> that touches app, infra or workflow paths, or manually. Change detection deploys only the backend, worker or frontend lanes that changed.

Aspect	Detail
Local	Developers run the frontend (Vite) and backend (Nest) locally against a local or dev database; a dev-auth bypass exists for local use only

Promotion path

feature/* branch → automatic deploy to **Dev** → merge to develop → release or hotfix PR to main (PR validation: lint, typecheck, tests, Terraform validate) → approved deploy to **Production**.

11 External dependencies

Systems outside this repository that the platform needs to work.

Dependency	What the platform needs from it	If it's down
Content Hub (MediaHub)	Catalog, KOL data, campaigns, analytics; Cognito M2M in both directions	Catalog serves cached data where available, otherwise empty states; report generation fails and can be retried
cht-reports	Report queue consumer, DynamoDB state table, S3 artifacts bucket and their resource policies	Reports stay queued; the rest of the platform is unaffected
cht-companion	Chat API over Service Connect	Chat unavailable; rest of the platform unaffected
Content Hub VTT ingest Lambda	Receives S3 notifications for Zoom transcripts	Transcripts not indexed; recordings still stored
Amazon Cognito	All user sign-in and M2M tokens	New sign-ins fail; existing sessions continue until expiry
Zoom, Stripe, Bill.com	Live sessions, attendance and payouts	Sessions or payouts delayed; webhooks retried by vendors; payment jobs held in SQS / DLQ
GitHub and GitHub Actions	Source, CI, deploys, Environment secrets	No deploys; running system unaffected
DNS and certificates	Route 53 zone for communityhealth.media , ACM certificates (us-east-1 and us-east-2)	Site unreachable if DNS fails or certificates lapse
Package registries and CDNs	npm, PyPI and Docker Hub at build time; Zoom CDN at runtime for the Meeting SDK	Builds fail; in-app Zoom falls back to the Zoom client link

12 Open items

To resolve

- **Reports in production.** The report table, queue and bucket are configured for dev only. Production values must be added to the platform Terraform variables before reports go live there.
- **MFA enforcement.** Cognito MFA is optional in both environments. Decide when to require it, starting with admins.
- **Stripe cutover.** Production still has Bill.com reminders enabled. Remove them after the Stripe cutover, and retire JotForm once legacy programs close.
- **DR compute.** The us-east-2 backend is deployed manually and API traffic is not routed there. Schedule and document a failover test.
- **Docs drift.** docs/engineering/architecture.md lists React 18, integrations.md still describes GoTrue auth, and deployment.md describes older production triggers. Update them to match this document.